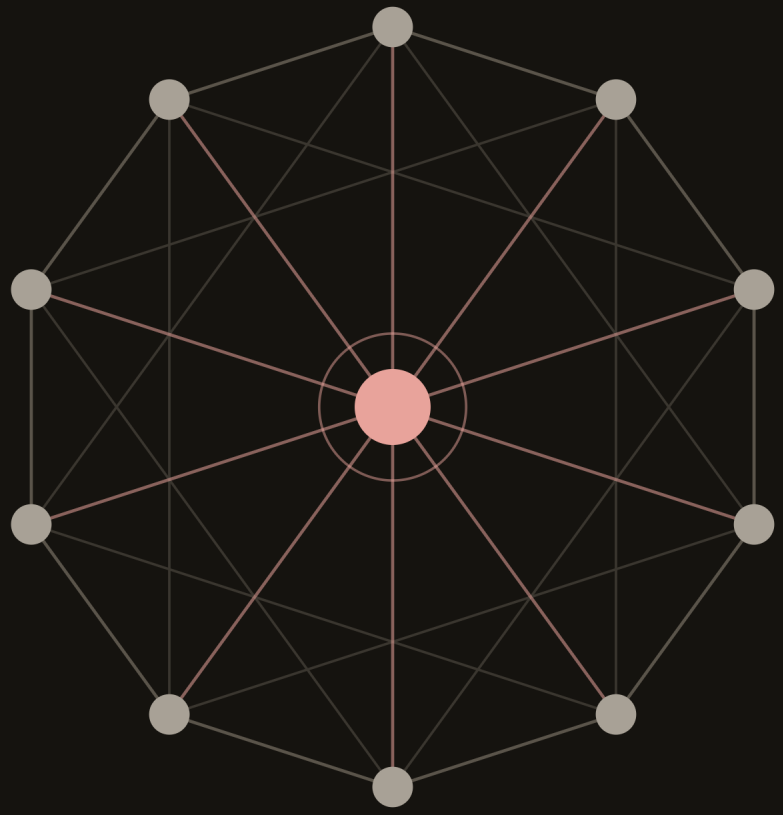


UMUNTU NGUMUNTU NGABANTU

I am because we are

Ubuntu, the African philosophy of shared humanity, and why it matters to the way software is built

ACHIM KIPKORIR LANGAT · COMPUTER SCIENCE

IN THIS PRESENTATION

Contents

Five chapters on an African idea that now runs through the software world.

01 **The philosophy**
What ubuntu means, and where it lives across Africa

02 **From philosophy to software**
Ubuntu Linux, and what makes software open source

03 **Ubuntu values in code**
How the philosophy maps onto the way software is built

04 **African builders**
Ushahidi, Masakhane, Zenzeleni and Kenya's developers

05 **Why it matters**
The impact on computing, and lessons for engineers

01 · THE PHILOSOPHY · A NGUNI PROVERB

“Umuntu ngumuntu ngabantu.”

A person is a person through other people.

ISIZULU · THE ISIXHOSA FORM IS UMNTU NGUMNTU NGABANTU



What ubuntu means

A way of seeing yourself through your relationships with other people.

01

Interdependence

No one exists in isolation. Who you are is shaped by the people around you.

02

Openness

“A person with ubuntu is open and available to others.” Archbishop Desmond Tutu, 1999.

03

Hospitality

Nelson Mandela's example: a traveller passing through a village is given food and water without having to ask.

04

Shared progress

Mandela added that ubuntu does not stop people working for themselves. The question is whether it lifts the community too.

Often summed up in English as “I am because we are” or “humanity towards others”.

Sources: Tutu, No Future Without Forgiveness, 1999; Wikipedia, “Ubuntu philosophy”

01 · THE PHILOSOPHY

One idea, many languages

ubuntu
isiZulu, isiXhosa · South Africa

botho
Sesotho, Setswana · Botswana, South Africa

utu
Kiswahili · Kenya, Tanzania

umunthu
Chichewa · Malawi, Zambia

hunhu
Shona · Zimbabwe

obuuntu
Luganda · Uganda

omundu
Luhya · Kenya; Herero · Namibia

kimuntu
Kikongo · Angola, DR Congo, Congo

In Kenya, the same spirit has a motto: harambee

Kiswahili for “all pull together”. Jomo Kenyatta made it a national rallying call after independence in 1963. It is on Kenya's coat of arms, and still the name for a community fundraiser.

Sources: Wikipedia, “Ubuntu philosophy” and “Harambee”

01 · THE PHILOSOPHY

Ubuntu in public life

From national mottos to constitutions, and finally into computing.

1963 — **Harambee in Kenya**

At independence, Jomo Kenyatta calls on the new nation to pull together.

1993 — **A constitutional value**

South Africa's interim constitution speaks of “a need for ubuntu but not for victimisation”.

1999 — **Tutu explains it to the world**

Desmond Tutu's No Future Without Forgiveness makes ubuntu central to the story of reconciliation.

2004 — **Into computing**

A new Linux operating system is released under the name Ubuntu.

Sources: Interim Constitution of South Africa, 1993, Epilogue; Tutu, 1999; Ubuntu.com

Ubuntu, the operating system

FIRST RELEASE

Ubuntu 4.10

“Warty Warthog”

Announced 20 October 2004

The launch announcement described ubuntu as an ancient African word for “humanity towards others”, and said the name suited a community project.

Who made it

South African entrepreneur Mark Shuttleworth and a small group of Debian developers, who founded the company Canonical.

What it is

An easy-to-use version of Linux, built on the Debian GNU/Linux distribution.

The promise

Free of charge, a new release every six months, and free security updates.

Who builds it

Canonical, other companies and thousands of volunteers share the work.

Sources: Ubuntu.com, “About”; Ubuntu 4.10 release announcement, October 2004

UBUNTU TODAY

9.5%

of all websites whose operating system is known run on Ubuntu: roughly one in ten.

Source: W3Techs, web technology surveys, 14 October 2026



What makes software open source

The Open Source Definition sets ten conditions. Four carry the spirit of ubuntu most clearly.

01

Source code

The program comes with its source code, so anyone can read it and learn from it.

02

Free redistribution

Anyone may pass the software on to others without paying a royalty for it.

03

Derived works

People may change it and share their improved versions under the same terms.

04

No discrimination

The licence may not exclude any person or group from using it.

In short: the code belongs to everyone who uses it, and everyone is welcome to make it better.

Source: Open Source Initiative, The Open Source Definition

03 · UBUNTU VALUES IN CODE

Values that line up

Ubuntu value	Open-source practice	Example
A person is a person through other people	Every project is built on code that other people wrote and shared	Ubuntu itself is built on Debian and Linux
Being open and available to others	Source code that anyone can read, study and learn from	The “Source code” condition of the Open Source Definition
Giving without being asked	Software that anyone may copy, use and pass on	Ubuntu has been free of charge since 2004
Respect and consensus	Communities agree in writing how members treat one another	The Ubuntu Code of Conduct
No one left outside the village	Licences may not shut out any person or group	“No discrimination against persons or groups”

Ubuntu written into the rules

- **Be considerate** · of users and colleagues
- **Be respectful** · and assume good intentions
- **Take responsibility** · for your words and actions
- **Be collaborative** · work in the open with others
- **Value consensus** · resolve disagreements
- **Ask for help** · and answer others helpfully
- **Step down considerately** · hand over your work

Humanity comes first

The code opens by saying that ubuntu is about showing humanity to one another.

Leadership as service

Leaders put the team's success first, give credit to others, and step aside when they have a conflict of interest.

Anyone can lead

Leadership comes from taking action, and lasts only while the community supports it.

Source: Ubuntu Code of Conduct, version 2.0, ubuntu.com

03 · UBUNTU VALUES IN CODE

A program is a program through other programmers.

The proverb, rewritten for our field: no developer builds alone.



Ushahidi, made in Kenya

Built in January 2008, in the crisis that followed Kenya's disputed 2007 election.

The idea

Ordinary people sent eyewitness reports of violence by text message and email, and each report was placed on a map.

The people

Ory Okolloh proposed it and built it with Erik Hersman, Juliana Rotich and David Kobia. Ushahidi is Kiswahili for “testimony”.

Open source

Its code is public, so any community can run its own copy and help improve it.

WHERE IT WENT NEXT

~40,000

reports received after the Haiti earthquake, 2010

8,000+

reports during Kenya's 2013 election, through the Uchaguzi project

125,000+

deployments in more than 160 countries by 2017, many for testing or training

Source: Wikipedia, “Ushahidi”

Names that say ubuntu

Masakhane

isiZulu: "we build together"

WHAT	A grassroots research community building language technology, such as machine translation, for African languages, led by Africans.
WHY	Africa has over 2,000 languages, and many get little attention from language technology.
WHEN	First presented at the AfricaNLP workshop at ICLR 2020, in Addis Ababa.

Zenzeleni

isiXhosa: "do it yourself"

WHAT	A solar-powered wireless network in Mankosi, a rural village in South Africa's Eastern Cape, owned and run by a local cooperative.
WHY	Data costs 20 to 40 times less than from major operators, and the money stays in the village.
WHEN	Started in 2012, with researchers from the University of the Western Cape.

Sources: masakhane.io; Orife et al., 2020; The Conversation, on Zenzeleni

04 · AFRICAN BUILDERS · KENYA

666,000

GitHub developer accounts in Kenya at the start of 2026: about 12 for every 1,000 people.

Across Africa: about 9.3 million.



Source: OpenUK, Open Source Africa Report 2026, using GitHub Innovation Graph data for Q1 2026

Why ubuntu matters to computing

01

Collaboration at scale

Shared ownership lets companies and thousands of volunteers improve one system, as they do with Ubuntu.

02

Access for everyone

Free software and community networks put computing within reach of people who could not otherwise afford it.

03

Local ownership

Communities build and run their own technology, from Mankosi's network to Masakhane's language tools.

04

Healthier teams

Codes of conduct turn respect, responsibility and consensus into everyday working rules.

05

An ethic for AI

Sabelo Mhlambi (Harvard, 2020) proposes ubuntu's relational view of the person as a framework for governing AI.

06

Africa's place in the field

Ubuntu, Ushahidi and Masakhane show African ideas and builders shaping global technology.

Lessons for engineers

01

Build on others, then give back

Use shared code with gratitude, and return your fixes and improvements to the projects you rely on.

02

Write for the next person

Readable code and clear documentation are how a developer stays open and available to others.

03

Build with the community you serve

Ushahidi, Masakhane and Zenzeleni worked because the people who needed them helped build them.

04

Measure success by who it lifts

Personal achievement is welcome. Ubuntu simply asks whether your work also strengthens the people around you.



CONCLUSION · UBUNTU SUMMIT 2024, TWENTY YEARS ON

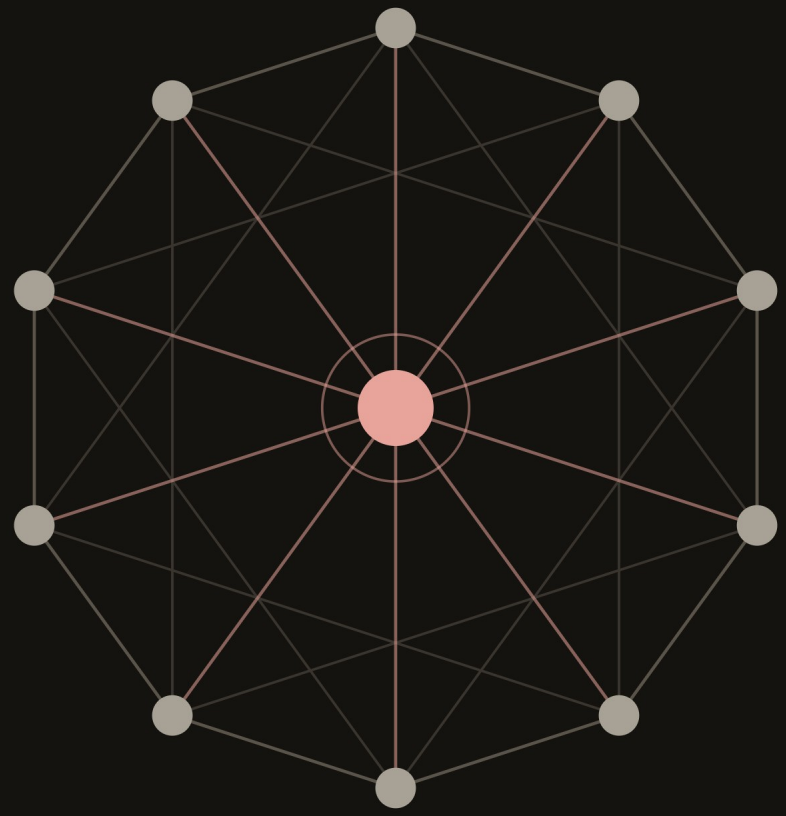
The best software, like the best community, is built through other people.

An African way of living together turns out to be a good description of how open-source software gets made.

Thank you.

Questions?

ACHIM KIPKORIR LANGAT · COMPUTER SCIENCE



BACK MATTER

Sources

THE PHILOSOPHY

- Tutu, D. No Future Without Forgiveness, 1999.
- Interim Constitution of the Republic of South Africa, 1993, Epilogue.
- Wikipedia. Ubuntu philosophy.
- Wikipedia. Harambee.
- Mhlambi, S. From Rationality to Relationality. Carr Center, Harvard, 2020.

UBUNTU AND OPEN SOURCE

- Ubuntu. About Ubuntu.
- Ubuntu 4.10 release announcement, 20 October 2004.
- Ubuntu. Code of Conduct v2.0.
- Open Source Initiative. The Open Source Definition.
- W3Techs. Usage statistics of Ubuntu, 14 October 2026.

AFRICAN BUILDERS

- Wikipedia. Ushahidi.
- Orife, I. et al. Masakhane: Machine Translation for Africa. 2020.
- Masakhane, masakhane.io.
- The Conversation. How a rural community built South Africa's first co-operative ISP.
- OpenUK and OpenHQ. Open Source Africa Report, 2026.

Photograph, slide 18: Ubuntu Summit 2024 group photo, 20 years of Ubuntu. [Photographer / source]